AQ2.5: Activity Questions 5- Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point
Match the matrices in Column A with their row operation steps (in the exact sequence given) in Column B, and their corresponding reduced row Echelon forms in Column C of Table M2W2AQ2.

Find the correct option.
☐ Applying (b) on (i) gives (2).
☐ Applying (a) on (ii) gives (1).
☐ Applying (a) on (i) gives (2).
☐ Applying (b) on (ii) gives (1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
Applying (a) on (i) gives (2).
Applying (b) on (ii) gives (1).

1 point
Choose the correct set of options.
☐
If two matrices AA and BB have the same reduced row echelon form, then AA must be equal to BB.
☐
If two matrices AA and BB have the same row echelon form, then AA must be equal to BB.
☐ The reduced row echelom form of a diagonal matrix with non-zero diagonal entries must be the identity matrix.
☐ The reduced row echelon form of a non-zero scalar matrix must be the identity matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The reduced row echelom form of a diagonal matrix with non-zero diagonal entries must be the identity matrix.
The reduced row echelon form of a non-zero scalar matrix must be the identity matrix.

(Answer questions 3 and 4 based on the below information.)
The three different types of elementary row operations that can be performed on a matrix are:

- Type 1:Type 1: Interchanging two rows.
- Type 2:Type 2: Multiplying a row by some constant.
- Type 3:Type 3: Adding a scalar multiple of a row to another row

1 point
Consider the four matrices given below:

$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -1 & 1 \\ 0 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & -1 & 1 \\ 0 & -1 & 1 \\ 1 & -3 & 3 \end{bmatrix}$.

Choose the set of correct options.

- [ ] Matrix B is obtained from matrix A by an elementary row operation of Type 1.
- [ ] Matrix C is obtained from matrix A by an elementary row operation of Type 1.
- [ ] Matrix D is obtained from matrix C by an elementary row operation of Type 3.
- [ ] Matrix A is obtained from matrix C by an elementary row operation of Type 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Matrix B is obtained from matrix A by an elementary row operation of Type 1.
Matrix D is obtained from matrix C by an elementary row operation of Type 3.
Matrix A is obtained from matrix C by an elementary row operation of Type 2.

1 point

Let A and B be square matrices of order 3. Consider the three equations below.

- Equation 1: $\det(A) = -\det(B)$
- Equation 2: $\det(A) = -c \det(B), c \in \mathbb{R}$
- Equation 3: $\det(A) = \det(B)$

Choose the set of correct options.

- [ ] If matrix B is obtained from matrix A by an elementary row operation of type 1, then equation 1 is satisfied.
- [ ] If matrix B is obtained from matrix A by an elementary row operation of type 1 followed by an elementary operation of type 2, then equation 2 is satisfied for some c.
- [ ] If matrix B is obtained from A by an elementary row operation of type 2, then equation 3 is satisfied.
- [ ] If matrix B is obtained from A by an elementary row operation of type 3, then equation 3 is satisfied.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If matrix B is obtained from matrix A by an elementary row operation of type 1, then equation 1 is satisfied.
If matrix B is obtained from matrix A by an elementary row operation of type 1 followed by an elementary operation of type 2, then equation 2 is satisfied for some c.
If matrix B is obtained from A by an elementary row operation of type 3, then equation 3 is satisfied.

1 point

Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ be a square matrix of order 3. Which of the statements below are true for matrix A?

- [ ] A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.
- [ ] The reduced row echelon form of matrix A is $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- [ ] The reduced row echelon form of matrix A is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- [ ] A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.
- [ ] A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.

No, the answer is incorrect.
Score: 0
Accepted Answers:

A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.

The reduced row echelon form of matrix A is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.

A can be transformed via elementary row operations into the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is in row echelon form.

Level 2:

Let the reduced row echelon form of a matrix A be $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$.

Suppose the first and third columns of A are $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ respectively. If the second column of the matrix A is given by $\begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$, then what is the value of $m_1 + m_2$.

[Hint: Start with an arbitrary column $\begin{bmatrix} a \\ b \end{bmatrix}$ as the second column of A.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Let the different types of elementary row operations be defined as follows:

- Type 1: Interchanging two rows.
- Type 2: Multiplying a row by some constant.
- Type 3: Adding a scalar multiple of a row to another row.

Which of the following statements are true?

If matrix A is obtained from matrix B by a finite number of elementary row operations, then matrix B can also be obtained from matrix A by a finite number of elementary row operations.
The reduced row echelon form of a matrix cannot be the identity matrix.
An upper triangular matrix, with value of all the diagonal elements equal to 1, is in row echelon form.
Identity matrix is in reduced row echelon form.
The reduced row echelon form of a scalar matrix (other than identity matrix) can be obtained by applying only elementary row operations of type 1.
The reduced row echelon form of a diagonal matrix (other than identity matrix) can be obtained by applying only elementary row operations of type 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If matrix A is obtained from matrix B by a finite number of elementary row operations, then matrix B can also be obtained from matrix A by a finite number of elementary row operations.
An upper triangular matrix, with value of all the diagonal elements equal to 1, is in row echelon form.
Identity matrix is in reduced row echelon form.

1 point

Let AA be a $3 \times 3$3 × 3 real matrix whose sum of entries of each column is 55 and sum of first two elements of each column is 33. Which of the following statements is (are) true?

[Hint: Row operation: adding one row to other row.]

- [ ] The determinant of matrix AA is a multiple of 55.
- [ ] The determinant of matrix AA is a multiple of 33.
- [ ] The determinant of matrix AA is a multiple of 1515.
- [ ] The determinant of matrix AA is a multiple of 22.
- [ ] The determinant of matrix AA is a multiple of 88.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The determinant of matrix AA is a multiple of 55.
The determinant of matrix AA is a multiple of 33.
The determinant of matrix AA is a multiple of 1515.
The determinant of matrix AA is a multiple of 22.
The determinant of matrix AA is a multiple of 88.

Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 2 \\ 1 & -1 & -1 \end{bmatrix}$A = ⎣01110−132−1⎦ be a matrix (where the first row denotes the top most row, and the ordering of the rows is from top to bottom). Consider the system of linear equations given by $Ax = b$Ax = b where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = ⎣x1x2x3⎦, and $b = \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix}$b = ⎣43−1⎦. Answer the following questions.

If RR be the reduced row echelon form of AA, then find the number of non-zero rows of RR.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

If $x_1 = 0$x1 = 0 is given, then find out the value of x_2x2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5